Chapter 5
Measurement Uncertainty in Decision Making



5.1 Introduction

Measurements are used to make important decisions ranging from accepting or rejecting product (Smith 1990) to deciding on R&D proposals (Cividino et al. 2019) to assessing the outcome of medical diagnostics (Bhise et al. 2018). However, because all measurements have an associated uncertainty, there is inherent risk that incorrect decisions will be made. Decisions regarding the air pressure of footballs (Sect. 1.4) may be of minor consequence, but decisions based on the pandemic fatality rate (Sect. 1.5) can have profound consequences. In cases like these, failure to address measurement uncertainty leaves the risk of incorrect decisions unknown and unmanageable.

This chapter provides a statistical framework for using measurement uncertainty to quantify risk in measurement related decisions. Test uncertainty ratios, measurement decisions, false accept and false reject probabilities, guardbanding, and risk associated with biased measurements are covered. These topics will first be presented in the context of a manufacturing environment in which the decision is to accept or reject manufactured product. The same topics will then be discussed in the context of a calibration lab in which a decision must be made regarding the state of a unit under test (UUT).

5.2 Measurement Uncertainty and Risk

In metrology, risk is defined as the probability that a measurement result will lead to an incorrect decision. Since the actual value of a measurand can never be known, there is always some probability that a measured value falls within an acceptable region when the actual value lies outside that region. This results in a *False Accept* (FA) condition. There is also some probability that a measured value falls outside an

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acceptable region when the actual value lies inside that region. This results in a *False Reject* (FR) condition. National standard ANSI/NCSLI Z540.3 2006 (see Sect. 3.7.4) recommends that compliance testing should control the false accept risk to be no more than 2%.

False accept and false reject risks are only one part of a full risk assessment. Another factor in the assessment, not considered in detail here, is the *consequence* of making an incorrect decision. These consequences may include the cost of rejecting a good part (inflating rework and scrap costs) or the cost of accepting a bad part (inflating future costs and risks). Depending on the consequences of an incorrect decision, false accept and false reject risks may need to be controlled at less than the typical 2%. However, controlling these risks at lower levels comes with its own cost, as more sampling, testing, and scrapping of marginal parts may be required.

5.2.1 Measurement Uncertainty and Risk in Manufacturing

In a manufacturing setting, a measurement is performed on each part (or a suitable sample of parts) at product acceptance to determine whether it meets requirements. Ideally, all parts that meet specifications are accepted and all parts that do not meet specifications are rejected. In the presence of measurement uncertainty, this ideal outcome is not possible. In the sections that follow, the focus is on the quantification and management of risks associated with uncertainty in the measurement of manufactured parts.

5.2.1.1 Test Uncertainty Ratio

An accept or reject decision is usually based on whether a part's measurement falls within its specification limits (*SL*). For example, if the voltage of a battery must fall within 10.0 V ± 1.0 V for its intended application, then the specification limit is $SL = 1.0$ V. Absolute limits may also be used: in this case the lower limit is $SL_L = 9.0$ V and the upper limit is $SL_U = 11.0$. Most equations in this chapter use absolute limits to avoid ambiguity. The JCGM (2012) uses the term "tolerance limits" in place of "specification limits," and defines the tolerance interval to be the range of acceptable values between the limits. Tolerance limits and specification limits are used interchangeably in this chapter.

Historically, the Test Uncertainty Ratio (TUR) has been used as an indicator of a measurement's suitability to make an accept or reject decision. The TUR is defined as TUR = [±Specification Limit]/[±Measurement Uncertainty at 95% confidence]. For example, if a part's length must be within ± 0.1 inch, and the caliper used to measure this length has total uncertainty of ±0.005 inch, the TUR is (0.1 inch/0.005 inch) = 20. When the specification limits are defined in terms of absolute limits, the TUR formula can be expressed as

$$\text{TUR} = \frac{SL_U - SL_L}{2U_c}, \tag{5.1}$$

where U_c is the expanded uncertainty (see Sect. 6.2.5), evaluated with a 95% confidence level.

Requiring a TUR of at least 4 (referred to as "meeting a 4:1 ratio") is often used to ensure the adequacy of a measurement to make a particular accept/reject decision, although sometimes a TUR of at least 10 is required for more conservative work. There are, however, important assumptions associated with using TUR as a metric and the requirement of a TUR of 4 or 10. Using a TUR assumes that all measurement biases have been removed from the measurement process and the measurements involved follow a normal distribution. If there are significant biases that cannot be removed, the TUR will not account for the increased risk (see Sect. 5.2.1.5). The TUR was originally introduced in the 1950s as a shorthand way to evaluate a measurement process without a full assessment of false accept and false reject probabilities (Harben and Reese 2012). With modern computers now readily available to evaluate these probabilities, the use of TUR alone becomes questionable. As a result, computing the actual risk is the recommended way to ensure false accept and false reject rates are acceptable. All of the risk computations presented in this chapter are supported by the uncertainty calculator that is available at https://sandiapsl.github.io/.

The TUR is still commonly used to determine if a measurement is adequate or if guardbanding (Sect. 5.2.1.4) is necessary to reduce the risk of false accepts. Within the NSE production agencies, whenever the TUR ratio is greater than or equal to 4:1, a measurement result can be directly compared to the specification limits for purposes of accept or reject. Otherwise, guardbanding is required (see Sect. 5.2.1.4).

Occasionally, the Test Accuracy Ratio (TAR) is used in place of the TUR. The TAR is similar to the TUR in its structure, but differs in that its denominator includes the uncertainty of only the test equipment, rather than including the additional Type A uncertainties, environmental effects, and any other known uncertainties in the measurement. The TAR can be expressed as:

$$\text{TAR} = \frac{\pm\text{Specification Limit}}{\pm\text{Accuracy of Test Equipment}}. \tag{5.2}$$

The TUR as an indicator of measurement capability is recommended as it includes all known uncertainties in the measurement. The TAR, while simpler to determine, could significantly understate the total uncertainty and associated risks.

5.2.1.2 Measurement Decisions

Any measurement performed on a manufactured part will have uncertainty, and any decision based on that measurement has inherent risk that should be considered. Figure 5.1 illustrates this concept, with y representing the measured

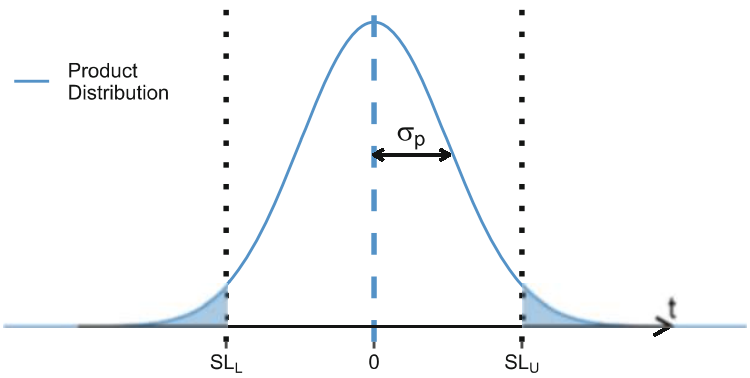


Fig. 5.3 Product distribution with specification limits. Dotted lines are shown at the specification limits

5.2.1.3 False Accept and False Reject Risks

The overall false accept and false reject risks associated with a manufacturing process can be determined by jointly considering the product and measurement uncertainty distributions (Smith 1990). The unconditional probability of false accept (*PFA*), sometimes called consumer's risk, is the probability that the measurement y is in tolerance ($SL_L < y < SL_U$) when the actual value t is out of tolerance ($t < SL_L$ or $t > SL_U$). This probability can be expressed as:

$$\begin{aligned} PFA = {} & Pr(SL_L \leq y \leq SL_U \text{ and } t < SL_L) \\ & + Pr(SL_L \leq y \leq SL_U \text{ and } t > SL_U). \end{aligned} \tag{5.6}$$

Assuming normal distributions, this probability can be calculated as:

$$PFA = \int_{-\infty}^{SL_L} \left(\int_{SL_L}^{SL_U} \frac{1}{\sigma_m \sqrt{2\pi}} e^{-\frac{1}{2\sigma_m^2}(y-t)^2} dy \right) \frac{1}{\sigma_p \sqrt{2\pi}} e^{-\frac{1}{2\sigma_p^2}\left(t-\mu_p\right)^2} dt +$$

$$\int_{SL_U}^{\infty} \left(\int_{SL_L}^{SL_U} \frac{1}{\sigma_m \sqrt{2\pi}} e^{-\frac{1}{2\sigma_m^2}(y-t)^2} dy \right) \frac{1}{\sigma_p \sqrt{2\pi}} e^{-\frac{1}{2\sigma_p^2}\left(t-\mu_p\right)^2} dt. \tag{5.7}$$

In this equation, the inner integral can be thought of as the conditional probability of accepting the part given a particular value of t. The outer integral averages over all values of t that result in the false accept condition.

Rather than requiring specific knowledge of σ_m, σ_p, and (SL_L, SL_U), it can be useful to express the *PFA* (Eq. (5.7)) in terms of TUR and *itp* using Eq. (5.1) and an estimate of σ_p based on *itp* and (SL_L, SL_U) (see Eq. (5.20)).

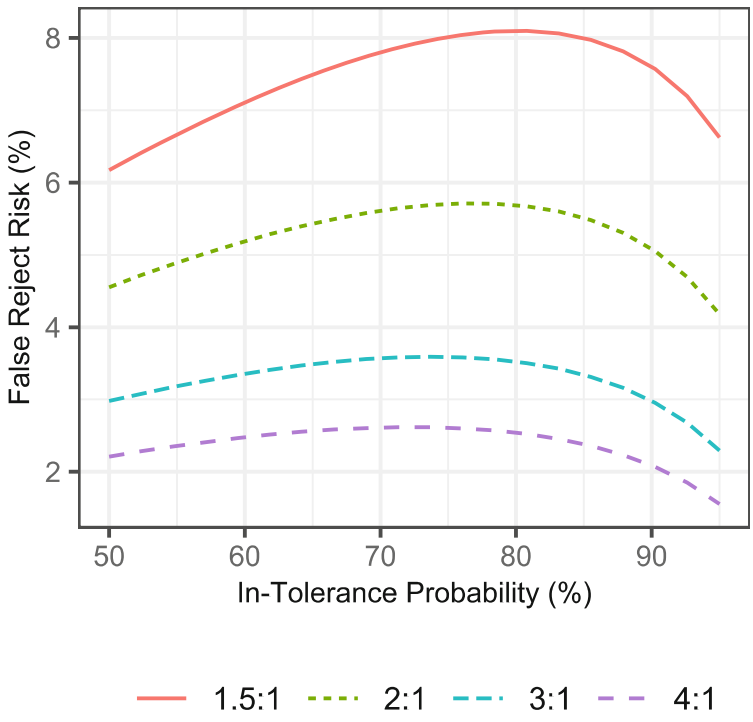


Fig. 5.5 False reject risk curve against in-tolerance probability. The curves represent various TURs

5.2.1.4 Guardbanding

Guardbanding modifies the specification limits to reduce the level of risk (*PFA*) associated with measurement uncertainty. The original specification limit, *SL*, is reduced by the guardband factor (GBF):

$$AL = SL \cdot \text{GBF} \tag{5.10}$$

to result in new *acceptance* limits *AL*. GBF is a multiplier between 0 and 1, with a value of 1 meaning no guardband is applied. In terms of absolute specification limits, absolute acceptance limits become

$$AL_L = SL_L + (1 - \text{GBF})\frac{SL_U - SL_L}{2} \tag{5.11}$$

$$AL_U = SL_U - (1 - \text{GBF})\frac{SL_U - SL_L}{2}. \tag{5.12}$$

The most popular method for calculating GBF is the root-sum-of-squares (RSS) method. The GBF for this method is

$$\text{GBF} = \sqrt{1 - \frac{1}{\text{TUR}^2}}. \tag{5.13}$$

between 10.055 V and 11.945 V. Risk for asymmetric requirements can be calculated in the manner described previously, with the region of integration modified for the asymmetric limits.

It should be pointed out that the need for guardbanding can be reduced by measuring marginal parts multiple times. Multiple independent measurements reduce the standard uncertainty, improving the ability to discriminate between good and bad parts. Cost models such as given by Smith (1990) consider the cost of additional measurements, along with the cost of false accept and false reject decisions.

5.2.1.5 Risk with Biased Measurements

While the 4:1 TUR requirement is commonly used to ensure a measurement is adequate for making an accept/reject determination, this metric assumes that the process distribution is centered between the specification limits, that is $\mu_p = (SL_U + SL_L)/2$. If this is not the case, TUR cannot be reliably used as an indicator of risk, however, the PFA and PFR equations are still valid assuming the correct μ_p is used.

The measurement uncertainty distribution is also assumed to be centered about the actual value t when calculating TUR. The measurement process is said to be biased if it is not centered about t and systematically overstates or understates the true value of the measurement. Properly accounting for measurement bias provides a more accurate risk evaluation. If bias is ignored, the risk might be understated, perhaps significantly.

In the presence of bias, the distribution of the measurement y, given the actual value t, shifts from a $N\left(t, \sigma_m^2\right)$ distribution to a $N\left(t - b_m, \sigma_m^2\right)$ distribution, where b_m is the measurement bias.

With bias b_m, the expressions for the PFA and PFR (without guardbanding) become

$$\begin{aligned} \text{PFA} = &\int_{-\infty}^{SL_L} \left(\int_{SL_L}^{SL_U} \frac{1}{\sigma_m\sqrt{2\pi}} e^{-\frac{1}{2\sigma_m^2}(y-(t-b_m))^2} dy \right) \frac{1}{\sigma_p\sqrt{2\pi}} e^{-\frac{1}{2\sigma_p^2}\left(t-\mu_p\right)^2} dt \\ &+ \int_{SL_U}^{+\infty} \left(\int_{SL_L}^{SL_U} \frac{1}{\sigma_m\sqrt{2\pi}} e^{-\frac{1}{2\sigma_m^2}(y-(t-b_m))^2} dy \right) \frac{1}{\sigma_p\sqrt{2\pi}} e^{-\frac{1}{2\sigma_p^2}\left(t-\mu_p\right)^2} dt. \end{aligned} \tag{5.18}$$

$$\begin{aligned}\text{PFR} = &\int_{SL_L}^{SL_U}\left(\int_{-\infty}^{SL_L}\frac{1}{\sigma_m\sqrt{2\pi}}e^{-\frac{1}{2\sigma_m^2}(y-(t-b_m))^2}dy\right)\frac{1}{\sigma_p\sqrt{2\pi}}e^{-\frac{1}{2\sigma_p^2}\left(t-\mu_p\right)^2}dt \\ &+\int_{SL_L}^{SL_U}\left(\int_{SL_U}^{+\infty}\frac{1}{\sigma_m\sqrt{2\pi}}e^{-\frac{1}{2\sigma_m^2}(y-(t-b_m))^2}dy\right)\frac{1}{\sigma_p\sqrt{2\pi}}e^{-\frac{1}{2\sigma_p^2}\left(t-\mu_p\right)^2}dt.\end{aligned} \tag{5.19}$$

Figure 5.10 shows PFA and PFR when μ_p is shifted by 60% of the specification limit $\left(\frac{SL_U-SL_L}{2}\right)$ without guardbanding (GBF = 1), and with $b_m = 0$. Figure 5.11 shows PFA and PFR when the measurement is biased by 20% of the specification limit ($b_m = 0.2\left(\frac{SL_U-SL_L}{2}\right)$).

As discussed earlier, if either shift is ignored and the 4:1 TUR rule is blindly applied, then the true PFA may exceed the traditionally accepted risk rate of 2%, even with a 4:1 TUR. For this reason, if it is not feasible to correct for measurement bias, the risk for the measurement process with bias needs to be explicitly quantified and deemed acceptable.

5.2.2 *Measurement Uncertainty and Risk in Calibration*

Calibration labs frequently perform tolerance tests on measurement equipment to ensure the equipment meets its specifications. Tolerance tests result in a pass or fail determination regarding whether the M&TE is in tolerance. Evaluating the

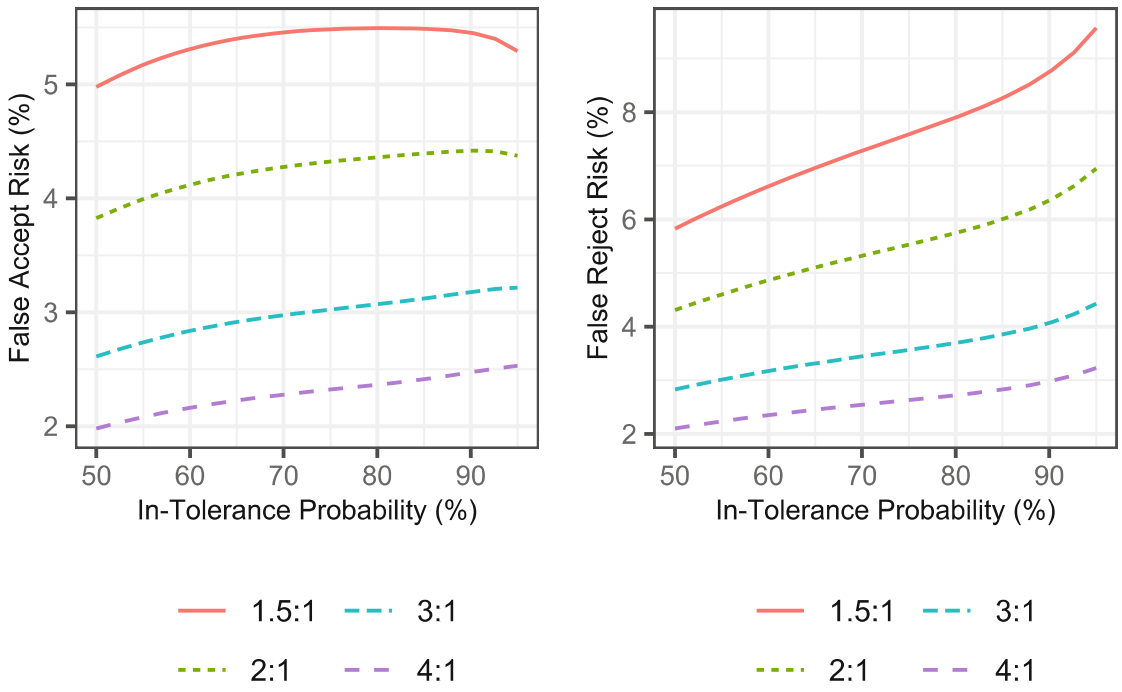


Fig. 5.10 False accept and reject with 60% process shift. The curves represent various TURs

$$\sigma_p \approx \frac{SL}{F^{-1}\left(\frac{1+itp}{2}\right)}. \tag{5.20}$$

The *itp* is again in-tolerance probability, sometimes referred to as end-of-period reliability (EOPR), based on the *observed* fraction of equivalent devices that were determined in-tolerance when submitted for recalibration. The function F^{-1} is the inverse normal cumulative distribution function, also known as the quantile function or percent-point-function. Most statistical software have functions to calculate this quantity, which assumes the population of equivalent M&TE follows a normal distribution centered at nominal.

When a calibration lab performs a true calibration, as opposed to a tolerance test (Sect. 2.2.4), no pass or fail determination is being made; only a value and measurement uncertainty are reported on the certificate. However, the calibration lab's accreditation statement lists a maximum uncertainty for a particular measurement, and the lab should verify the risk of exceeding their accredited uncertainty is minimal.

Finally, when a calibration lab makes an incorrect decision when testing M&TE, the consequences and costs of such a decision are different from a manufacturing setting. Depending on what the M&TE is used for, any measurement made with that equipment during its calibration cycle should be considered suspect. If the lab calibrates a reference standard for use in another calibration lab down the traceability chain, the incorrect decision can affect many downstream calibrations.

5.2.2.1 Decision Rules in Calibration

ISO 14253-1 2017 defines requirements for calibration labs, including a requirement to state a "decision rule" on calibration certificates. The decision rule defines the action(s) to take given a measurement result. "Simple acceptance," sometimes called "as-read," is the most common decision rule, where the outcome is a pass if the measurement as-read from the M&TE indicates a passing condition. This decision rule is commonly employed when the TUR is greater than 4:1. When the TUR is less than 4:1, a "guarded acceptance" decision rule is used, where the as-read measurement must be compared to a reduced acceptance interval.

Some calibration labs have begun using "indeterminate acceptance" decision rules, where a measured value within the acceptance limits is considered a pass, a value outside the specification limits is a fail, but measured values falling within the guardband (between the acceptance limit and specification limit) is an "indeterminate pass" condition. In this situation, the lab's measurement indicates a pass condition, but with a high risk of being OOT. "Indeterminate fail" conditions are also occasionally used, when the measured value is outside the specification limits, but only by some small tolerance. The indeterminate outcomes effectively shift the pass/fail decision on to the end-user of the equipment.

Sometimes, non-binary decision rules are used. For example, when manufacturing a part with a maximum diameter requirement of 1.0 cm, all components measuring less than 1.0 cm will be accepted, parts measuring greater than 1.2 cm will be rejected, but parts measuring between 1.0 cm and 1.2 cm can be sent for rework, and not rejected outright. The same concepts from this chapter apply to non-binary decision rules assuming care is taken to appropriately set the integration regions in the risk calculations.

5.3 Summary

Measurements are used to make conformity decisions, yet because all measurements have uncertainty, there is inherent uncertainty in the decision outcome and risk of making an incorrect decision. Historically, a TUR of at least 4:1 has been used to ensure a measurement process is adequate for ensuring compliance with a specification. When a 4:1 ratio cannot be achieved, guardbanding should be applied to reduce the risk of false accept decisions. However, guardbanding also increases the risk of false reject decisions. Any non-normal behavior or bias in the distributions will affect the false accept and reject levels and is not accounted for in the TUR 4:1 rule of thumb. In this case, a full analysis of risk, including evaluation of *PFA* and *PFR*, is the best option to ensure adequate *PFA* rates.

5.4 Related Reading

Additional details and more motivation for the risk calculations as applied to manufacturing are given by Smith (1990). He includes an evaluation of the expected cost per part based on the cost of a false accept, the cost of a false reject, and the cost of making additional measurements. Basnet and Case (1992) and Sankle and Singh (2012) discuss evaluation of manufacturing risk based on sampling from a batch process rather than testing every single part. The process capability index, *Cpk*, closely related to *itp* defined in this chapter, is a commonly used measure of the ability of a manufacturing process to meet requirements. As discussed in Mottonen et al. (2008), if the *Cpk* is high enough, risk is minimal regardless of the test measurement.

The discussion in this chapter calculates risk based on a single feature of a part. In many cases, a part will have numerous specifications that must all be met in order to meet requirements. Jackson (2019) discusses combined false accept rates and when multiple individual attributes are measured on one component.

Much of the background work on consumer's and producer's risk as applied to calibration was done by Deaver (1993, 1995) and Dobbert (2007, 2008). This work was incorporated into the NCSLI's guidance for application of standard Z540.3 in National Conference of Calibration and Standards Laboratories International (2009). Mimbs (2007) gives an overview of measurement risk with some interesting history

on the confusing inconsistent definitions of TAR and TUR and NASA's decision to change from a 10:1 to a 4:1 TUR requirement.

JCGM (2012) provides general statistical guidance in calculating risk as well as setting acceptance and tolerance limits. To avoid confusion between different and often ambiguous definitions of TUR and TAR found in literature, it uses the term "measurement capability index," the same quantity as TUR given in this chapter. ISO 10576-1 (2013) is the international standard, referenced by JCGM 106, on conformity assessment practices. ISO 14253-1 considers conformity assessment on geometrical product specifications. The NCSLI also offers recommended practices for applying the concepts in these standards with NCSLI RP-18 (2014), which includes a discussion of risk assessment using a Bayesian approach. Interpreting manufacturer's specifications with regard to risk assessment is discussed in National Conference of Calibration and Standards Laboratories International (2016). Finally, IEC 31010 2009 covers risk management from a general perspective, not specific to risk only in measurements.

Several other methods of setting acceptance limits to mitigate measurement risk have been proposed. Deaver (1994) compares several common methods based on TUR. Easterling et al. (1991) and Kim et al. (2007) consider total cost minimization methods, and Hund et al. (2017) proposes a data-driven probabilistic approach.

Mimbs (2011) discusses the usefulness of end-of-period reliability (EOPR) as a means for satisfying the 2% PFA metric required by Z540.3, differentiating observed EOPR from true EOPR. Mimbs, along with Dobbert (2008), introduce a Monte Carlo approach to calculating false accept and false reject risk that requires no calculus but does require a suitable random number generator (Chap. 8). Singh (1996) considered non-normal distributions in risk analysis, using an Edgeworth series expansion to approximate non-normal distributions. Castrup (2001) mentions the importance of accounting for bias in normal distributions, while Delker (2020) directly considers the effect of bias and non-normal behavior, as quantified by skewness and kurtosis, on false accept and false reject probabilities.

5.5 Exercises

1. Calculate the test uncertainty ratio (TUR) of a resistance measurement used to verify a resistor falls within requirements 1000 Ω ± 4 Ω, measured with an ohmmeter having ±1 Ω (95% confidence) uncertainty.
2. A caliper is used to measure a component and ensure it has a diameter of less than 0.5 cm. If the caliper has total (95% confidence) uncertainty of 0.1 mm, calculate an appropriate acceptance limit for this measurement.
3. A caliper is used to measure a component and ensure its width falls between 9.9 cm and 10.1 cm. The caliper has total (95% confidence) uncertainty of 0.05 mm.

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